

IBRAHIM MOHAMED IBRAHIM

Renewable Energy Engineer | PV System Design | Hybrid PV & BESS

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PROFESSIONAL SUMMARY

Renewable energy engineer with close to two years of hands-on experience designing and documenting PV systems, from residential rooftops to utility-scale plants. Proven track record in on-grid, off-grid, pumping, and hybrid PV+BESS/microgrid design, PVsyst simulation and energy-yield assessment, full CAD documentation, and tender/technical-proposal preparation. Combines strong technical design skills with commercial acumen (ROI, BOQs, vendor evaluation) to deliver bankable, cost-effective solar solutions.

CORE TECHNICAL SKILLS

- **PV System Design:** On-Grid, Off-Grid, Solar Pumping, and Hybrid Systems; PV array & string sizing; inverter selection; MPPT configuration; AC/DC system design
- **Hybrid PV & BESS:** AC-Coupled and DC-Coupled PV+BESS design; BESS sizing (DoD, SoH, Round-Trip Efficiency, C-Rate); PCS selection & sizing; microgrid architecture and load management
- **Electrical Design & Calculations:** Load assessment, DC/AC cable sizing, voltage-drop calculations, protection device selection & sizing, string voltage/current verification, MPPT allocation, earthing & grounding design
- **CAD & Engineering Drawings:** General PV layouts, string distribution layouts, concrete/pile layouts, wiring diagrams, general & detailed single-line diagrams (SLDs) with string-per-MPPT distribution, AC/DC cable routing, earthing layouts, shop drawings
- **Performance & Feasibility:** PVsyst simulation, energy yield assessment, performance ratio (PR) and loss analysis, shading analysis, technical feasibility studies, ROI and payback/cost analysis
- **Tendering & Technical Proposals:** Tender document & scope review, technical proposal preparation, BOQs & material take-offs, vendor quotation evaluation, technical compliance sheets & deviation statements, cost estimation, RFIs/clarifications, bid meeting coordination
- **Software:** PVsyst, PV*SOL, AutoCAD, SketchUp 3D, Microsoft Excel, MATLAB/Simulink, EPLAN, TIA Portal, Factory I/O
- **Additional Technical Exposure:** Classic control & PLC programming (Siemens, Schneider, Delta); electric motor & drive programming (INVT, Yaskawa, Delta); embedded systems & PCB design (ongoing)

PROFESSIONAL EXPERIENCE

Technical Office Engineer | Mashreq for Energy Systems | Cairo, Egypt

2025 – Present

- Designed PV systems using SketchUp, PVsyst, and AutoCAD, including string layouts, wiring diagrams, single-line diagrams (SLDs), and concrete/pile layouts, across residential, commercial, and utility-scale projects.
- Designed hybrid PV+BESS systems in both AC-coupled and DC-coupled configurations, including microgrid architecture using BESS and PCS.
- R&D: developed a powerful Excel-based automation tool for hybrid PV system design and sizing, streamlining calculations and reducing manual design time.
- Prepared technical and financial offers and ROI analyses for solar projects, and supported tender preparation: BOQs, technical compliance sheets, vendor quotation evaluation, and clarifications.
- Used Excel for system sizing, quantity schedules, and project cost evaluation.
- Coordinated with engineering, procurement, and project teams to optimize designs for efficiency, cost-effectiveness, and client requirements.

Technical Office Engineer — Internship | Mashreq for Energy Systems | Cairo, Egypt

Summer 2025

- Assisted in the design and documentation of PV projects; prepared shop drawings, as-built plans, and technical submissions.
- Supported technical studies for solar pumping systems and on-grid PV stations.

EDUCATION

Bachelor's Degree in Renewable Energy — Solar Energy Specialization

New Cairo Technological University (2022-2026)

High School Diploma

Modern School Ahmed Esmat (2021 – 2022)

SELECTED GRADUATION PROJECTS

Intelligent Photovoltaic Cleaner with IoT Integration — *Team Leader*

- Led development of an IoT-based automated PV panel cleaning system designed to minimize performance losses from dust and surface contamination.
- Coordinated the team across mechanical, electrical, control, and IoT subsystems, including DC motors, brush mechanisms, servo actuators, solenoid valves, and water circulation/filtration/pumping.
- Developed and integrated the Arduino-based control architecture with IoT monitoring and control.
- Led the project through multiple national innovation and technology competitions, achieving several competitive rankings.

Solar Collector & Dryer System Optimization — *Implementation Team Member*

- Participated in the implementation of a spherical (hemispherical) solar collector integrated with an agricultural drying system.
- Contributed to building the collector, drying chamber, airflow system, and control/monitoring components.
- Supported integration of temperature, humidity, and thermal measurement systems for monitoring collector and drying-system performance.
- Participated in testing, performance evaluation, and iterative optimization to improve heat distribution, drying uniformity, and thermal performance.

ACHIEVEMENTS & RECOGNITION

Solar Collector & Dryer System

- Secured project funding from the Academy of Scientific Research and Technology (ASRT).
- Received funding and technical support from the Innovation Support Foundation (ISF).
- Recognized through the iHub – Ain Shams University Innovation Hub program.
- 1st Place – Science & Innovation Competition (ASRT); 2nd Place – Technological Conference; 4th Place – ISEC Competition; 6th Place – Research Center Hackathon; Finalist – Hult Prize Competition.

Intelligent Photovoltaic Cleaner with IoT Integration

- 1st Place – Kemteck Valley Competition (Egyptian Technological Universities level).
- 2nd Place – Kemteck Valley Competition (New Cairo Technological University level).
- 3rd Place – Innovators Support Fund Competition (Greater Cairo Region).
- 10th Place – Electronic Research Institute Hackathon.
- Qualified through the Innovators Support Fund competition rounds for the Gen-Z TV Program.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Utility Scale PV Training Program – ECO Egypt (Aug–Sep 2025): solar basics, PV components, medium-voltage power plant components; design of on-grid, off-grid, pumping, and utility-scale PV stations; PVsyst/SketchUp/AutoCAD training; tendering, ROI, PV project & progress management, site visits. Certificate ID: EE-USPV-J25-150.
- Summer Camp Solar Training – LONGI (September 2024).
- ECO EGY Engineering Consulting Group – Level 1 (August 2023) & Level 2 (September 2023).
- HA Consulting Group for Automation Control – Basic Level (Jul–Sep 2024).
- Jelecom Supporting Academy: Microchip Basic Electronics & PCB Design; Microcontroller Basic Applications (Jul–Sep 2023).
- Software training: PVsyst, SketchUp 3D, MATLAB/Simulink, EPLAN, TIA Portal, Factory I/O, Proteus, Multisim, Logic X Pro.
- ICDL Certification (Microsoft Office Suite).

LANGUAGES

- Arabic — Native
- English — Intermediate (B1)
- German — Beginner (A1, in progress)